F_{IS}

2024-12-10

```
res_dir <- "../results/s08_FIS/"
```

Load gt

```
malagasy <- gt_load("../data/intermediate/rbind_data/malagasy_merge_clean.gt")
```

This script calculates F_{IS} in the Malagasy population, using the Malagasy only dataset containing 233 individuals and 3 loci.

Loci and sample count

```
nrow(malagasy)
```

```
## [1] 233
```

```
count_loci(malagasy)
```

```
## [1] 579742
```

Population level F_{IS}

F_{IS} was calculated first at the population level, using the method of (Weir and Goudet 2017).

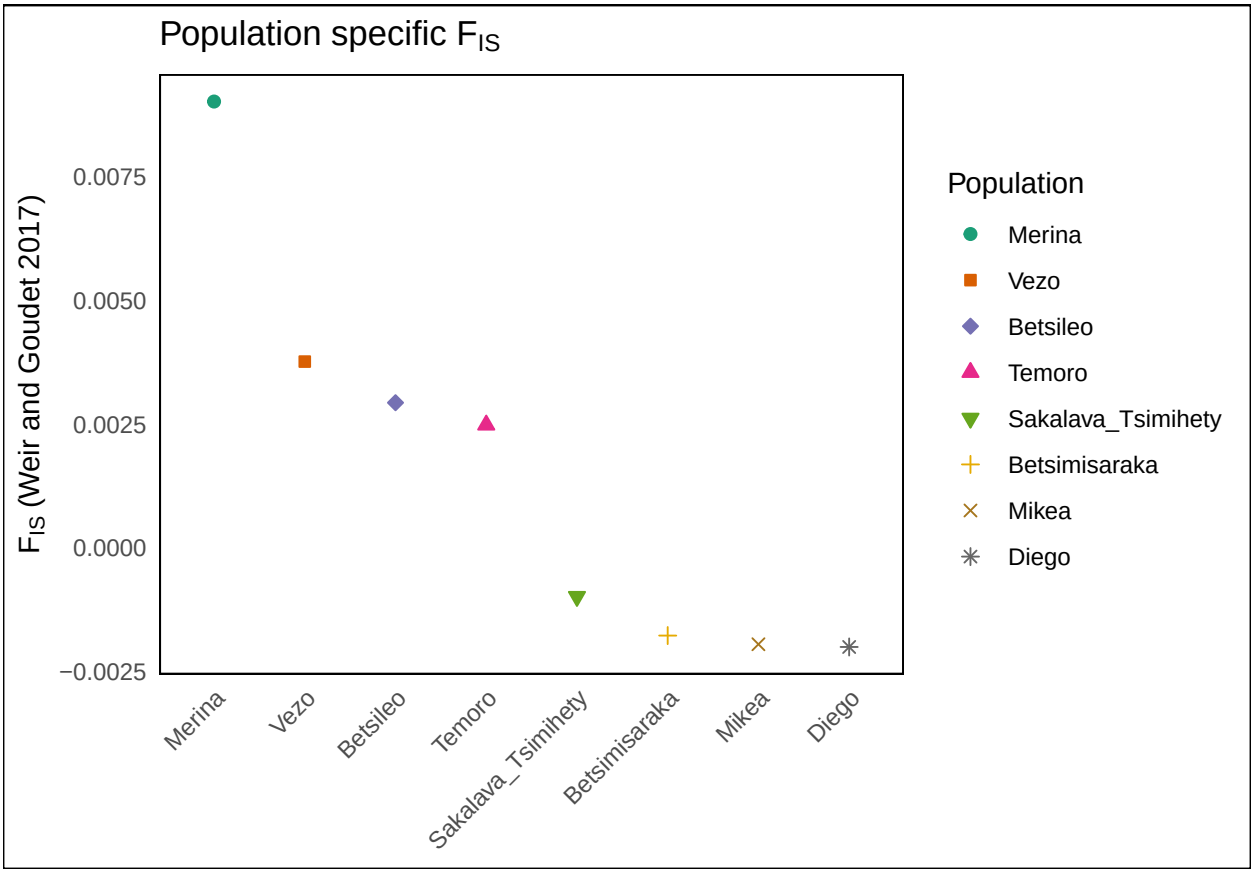
```
fis <- malagasy %>%
  group_by(population) %>%
  pop_fis(
    method = "WG17",
    by_locus = FALSE,
    include_global = FALSE,
    allele_sharing_mat = NULL
  )

# create dataframe for plotting
fis <- as.data.frame(fis)
fis$population <- rownames(fis)

# rearrange fis column fis in descending order
fis <- fis[order(fis$fis, decreasing = TRUE), ]

# Reorder the population variable as a factor based on the descending order
# of F~IS~
fis$population <- factor(fis$population, levels = fis$population)
```

```r
ggplot(fis, aes(
  x = population, y = fis, fill = population,
  colour = population, shape = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = expression("Population specific F"[IS]), x = NULL,
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)")
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank()) +
  scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```




```
ggsave(paste0(res_dir, "pop_fis.jpg"), width = 7, height = 7, dpi = 300)
```

This plot shows that population level F_{IS} is highest in the Merina population, with Diego, Mikea, and Betsimisaraka populations having similarly low F_{IS} values. The above 0 value for the Merina indicates an excess of homozygosity among the Merina, compared to neutral expectations.

- How does this comapre to the F_{IS} values for the other populations? Is it significantly different? And how can we test for this? See Rao and Inbaraj, 1977

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt")

# Add population metadata
pop_names <-
  read.csv("../data/raw/data_input/igsr-1000-genomes-phase-3-release.tsv",
    sep = "\t"
  )

# find matching id in pop_names Sample.name and mad_bor_g1000_sum id
# for the matches, replace the population with the population in pop_names
for (i in seq_len(nrow(pop_names))) {
  if (pop_names$Sample.name[i] %in% mad_bor_g1000_sum$id) {
    mad_bor_g1000_sum[mad_bor_g1000_sum$id == pop_names$Sample.name[i], "population"] <- pop_names$Popul
  }
}

# replace "Luhya,Luhya" with "Luhya"
```

```r
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Luhya,Luhya"] <- "Luhya"
# replace "Esan,Esan" with "Esan"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Esan,Esan"] <- "Esan"
# replace "Gambian Mandinka,Gambian" with "Gambian Mandinka"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Gambian Mandinka,Gambian"] <- "Gambian Ma
# replace "Mende,Mende" with "Mende"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Mende,Mende"] <- "Mende"
# replace "Kinh,Kinh Vietnamese" with "Kinh Vietnamese"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Kinh,Kinh Vietnamese"] <- "Kinh Vietnames
# replcae "Punjabi,Punjabi" with "Punjabi"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Punjabi,Punjabi"] <- "Punjabi"
# replcae "Benga,Bengali" with "Bengali"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Bengali,Bengali"] <- "Bengali"

# replace any _ with a space
mad_bor_g1000_sum$population <- gsub("_", " ", mad_bor_g1000_sum$population)
```

```r
fis_all_pops <- mad_bor_g1000_sum %>%
  group_by(population) %>%
  pop_fis(
    method = "WG17",
    by_locus = FALSE,
    include_global = FALSE,
    allele_sharing_mat = NULL
  )

# create dataframe for plotting
fis_all_pops <- as.data.frame(fis_all_pops)
fis_all_pops$population <- rownames(fis_all_pops)

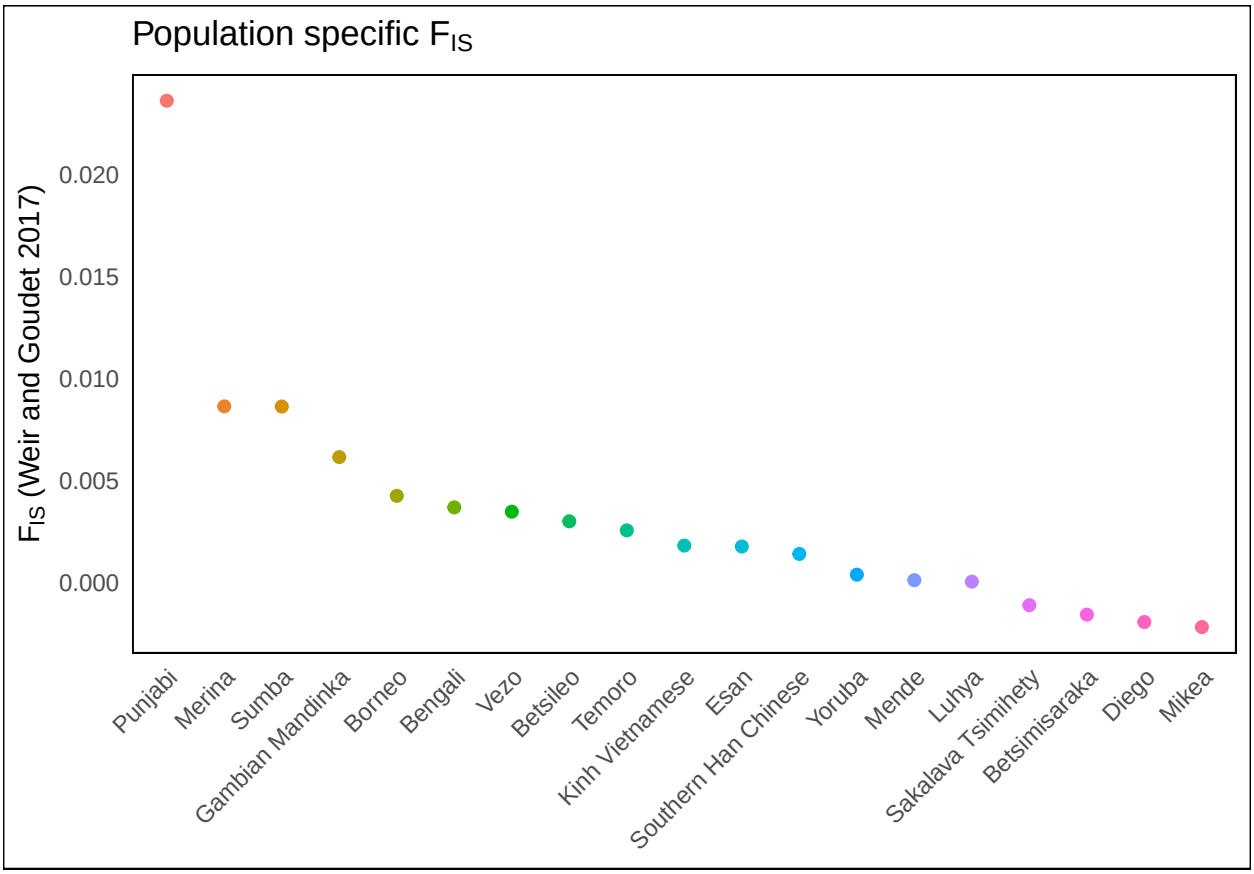
# rearrange fis_all_pops column fis_all_pops in descending order
fis_all_pops <-
  fis_all_pops[order(fis_all_pops$fis_all_pops, decreasing = TRUE), ]

# Reorder the population variable as a factor based on the descending
# order of F~IS~
fis_all_pops$population <- factor(fis_all_pops$population,
  levels = fis_all_pops$population
)

# Remove Timor
fis_all_pops <- fis_all_pops %>% filter(population != "Timor")

ggplot(fis_all_pops, aes(
  x = population, y = fis_all_pops,
  fill = population, colour = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = expression("Population specific F"[IS]), x = NULL,
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)")
  ) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank()) +
  # guides(
```

```
  #   fill = guide_legend(title = "Population"),
  #   colour = guide_legend(title = "Population"),
  #   shape = guide_legend(title = "Population")
  # ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```




```
ggsave(paste0(res_dir, "pop_fis_global_comparison.jpg"), width = 7, height = 7, dpi = 300)
```

When compared to the global populations in the dataset, Merina are second highest, but still far lower than the F_{IS} value for the Punjabi population

Population level F_{IS} for Merina subsets

```
# Load fastmixture Q matrix
qmats <- readRDS("../data/intermediate/fastmixture/fastmixture_m.RData")
```

```
#mad_bor_g1000_sum <-
#  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt")
```

```
# tidy with the gt
tidyq <- augment(get_q_matrix(qmats, k = 2, run = 1), data = mad_bor_g1000_sum)
```

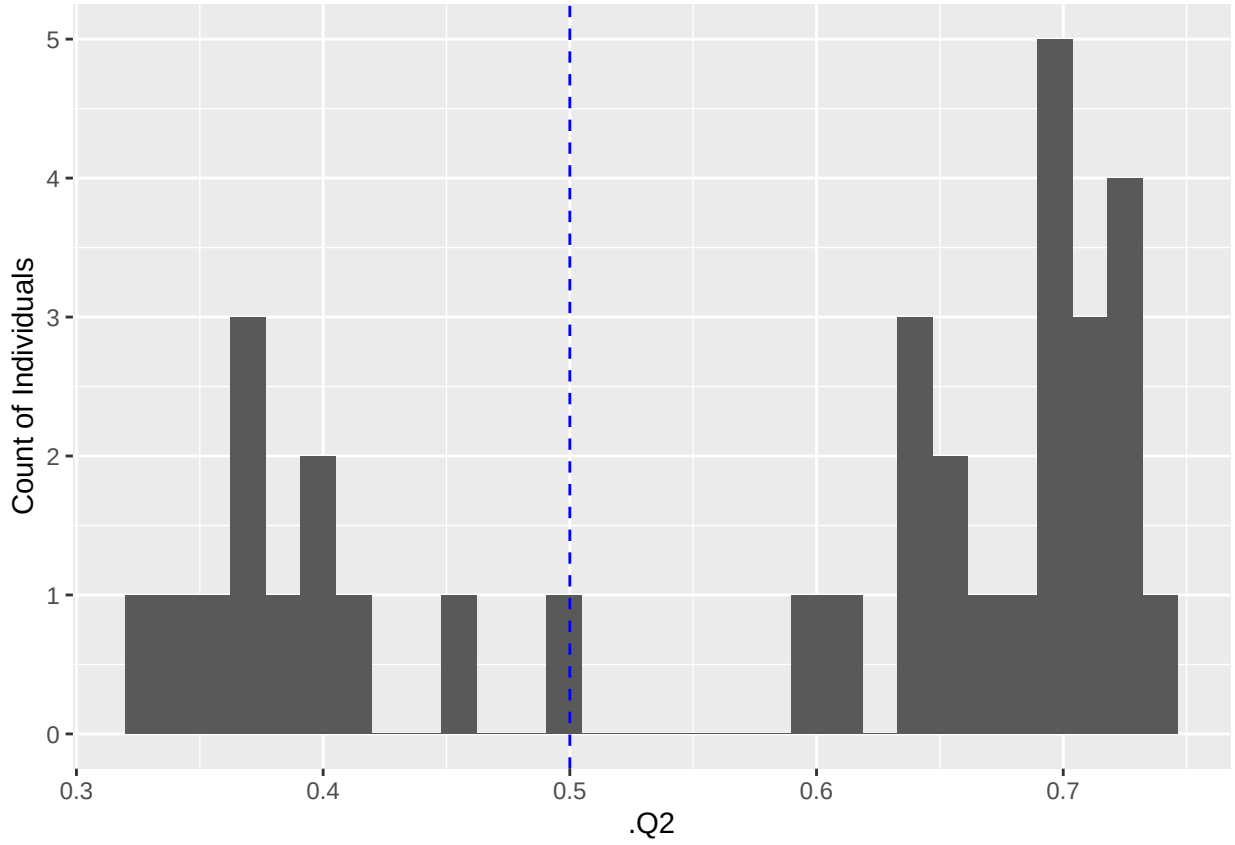
```
# subset to only malagasy
all_malagasy <- c(
  "Betsimisaraka", "Merina", "Sakalava_Tsimihety",
  "Betsileo", "Diego", "Vezo", "Mikea", "Temoro"
)

malagasyq <- tidyq %>% filter(population %in% all_malagasy)
```

```
# subset the malagasyq dataframe to only the "Merina"
merina <- malagasyq %>% filter(population == "Merina")

# plot column .Q3 for the merina
ggplot(merina, aes(x = .Q2)) +
  geom_histogram() +
  labs(y = "Count of Individuals") +
  geom_vline(xintercept = 0.5, linetype = "dashed", color = "blue")
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
ggsave(paste0(res_dir, "merina_q3.jpg"), width = 7, height = 7, dpi = 300)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

Separate the two Merina clusters according to ancestry proportion

```
# create column $group in merina - those with .Q3 higher than 0.5 are 1
# and those lower than 0.5 are 2
merina$group <- ifelse(merina$.Q2 > 0.5, "Social Group A: Merina", "Social Group B: Merina")
```

Calculate F_{IS} for Merina groups separately

F_{IS} was re-calculated, this time splitting the Merina into high and low African ancestry subsets.

```
fis_merina <- merina %>%
  group_by(group) %>%
  pop_fis(
    method = "WG17", by_locus = FALSE,
    include_global = FALSE, allele_sharing_mat = NULL
  )

fis_merina <- as.data.frame(fis_merina)
fis_merina$population <- rownames(fis_merina)
colnames(fis_merina) <- c("fis", "social_group")

fis_merina$population <- "Merina"
# reorder columns
fis_merina <- fis_merina[, c("fis", "population", "social_group")]

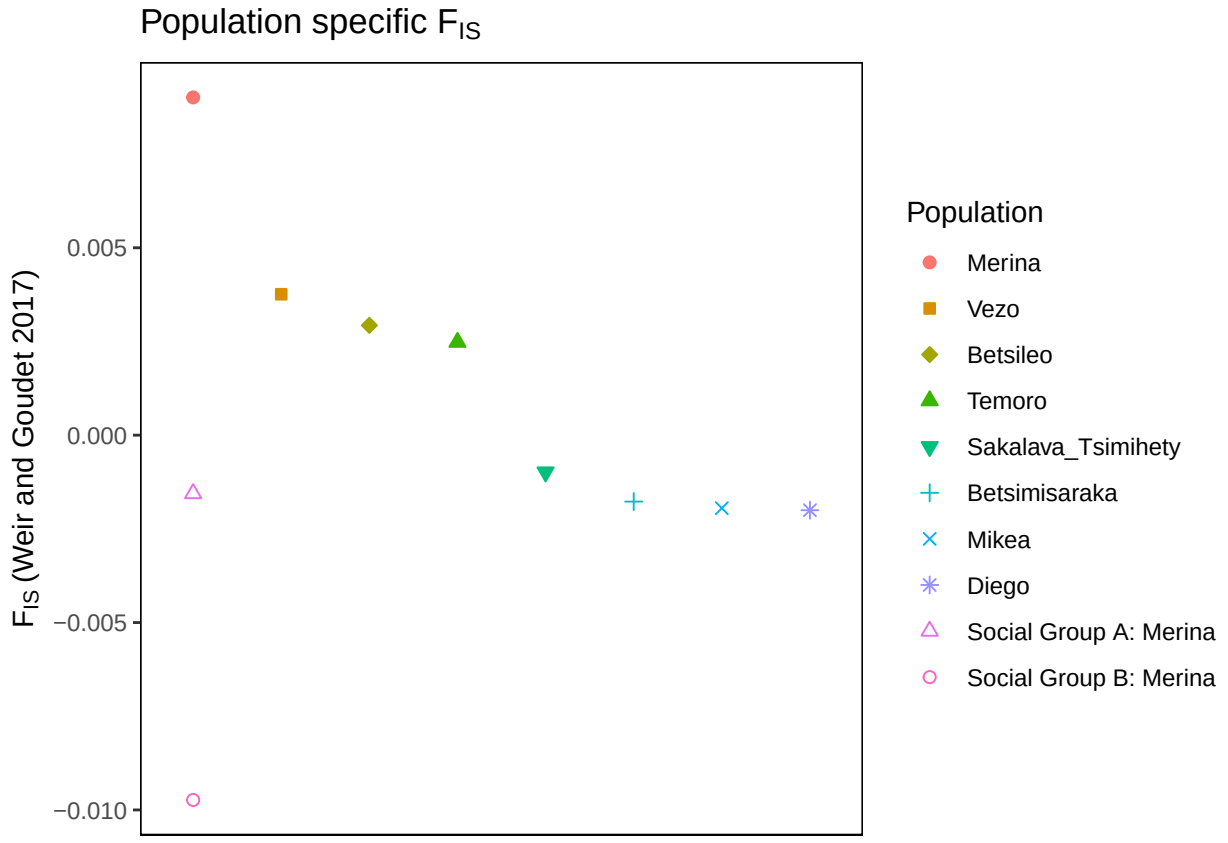
# add fis_merina to fis
fis$social_group <- fis$population
fis <- rbind(fis, fis_merina)


# rearrange fis column fis in descending order
fis <- fis[order(fis$fis, decreasing = TRUE), ]
```

Plot

```
ggplot(fis, aes(
  x = population, y = fis, fill = social_group,
  colour = social_group, shape = social_group
)) +
  geom_point(size = 1.8) +
  labs(
    title = expression("Population specific F"[IS]), x = NULL,
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)")
  ) +
  # scale_color_brewer(palette = "Dark2")+
```

```
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8, 2, 1)) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank()) +
  # scale_fill_brewer(palette = "Dark2") +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  )+
  theme(
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```




```
ggsave(paste0(res_dir, "pop_fis_social_groups.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
fis_all_pops <- fis_all_pops %>%
  mutate(social_group = population) %>%
  filter(!population %in% c(
    "Betsimisaraka", "Merina", "Diego",
    "Sakalava_Tsimihety", "Betsileo",
```

```
    "Vezo", "Mikea", "Temoro"
  ))
colnames(fis_all_pops) <- c("fis", "population", "social_group")
fis_all <- rbind(fis, fis_all_pops)
# rearrange fis_all_pops column fis_all_pops in descending order
fis_all <-
  fis_all[order(fis_all$fis, decreasing = TRUE), ]

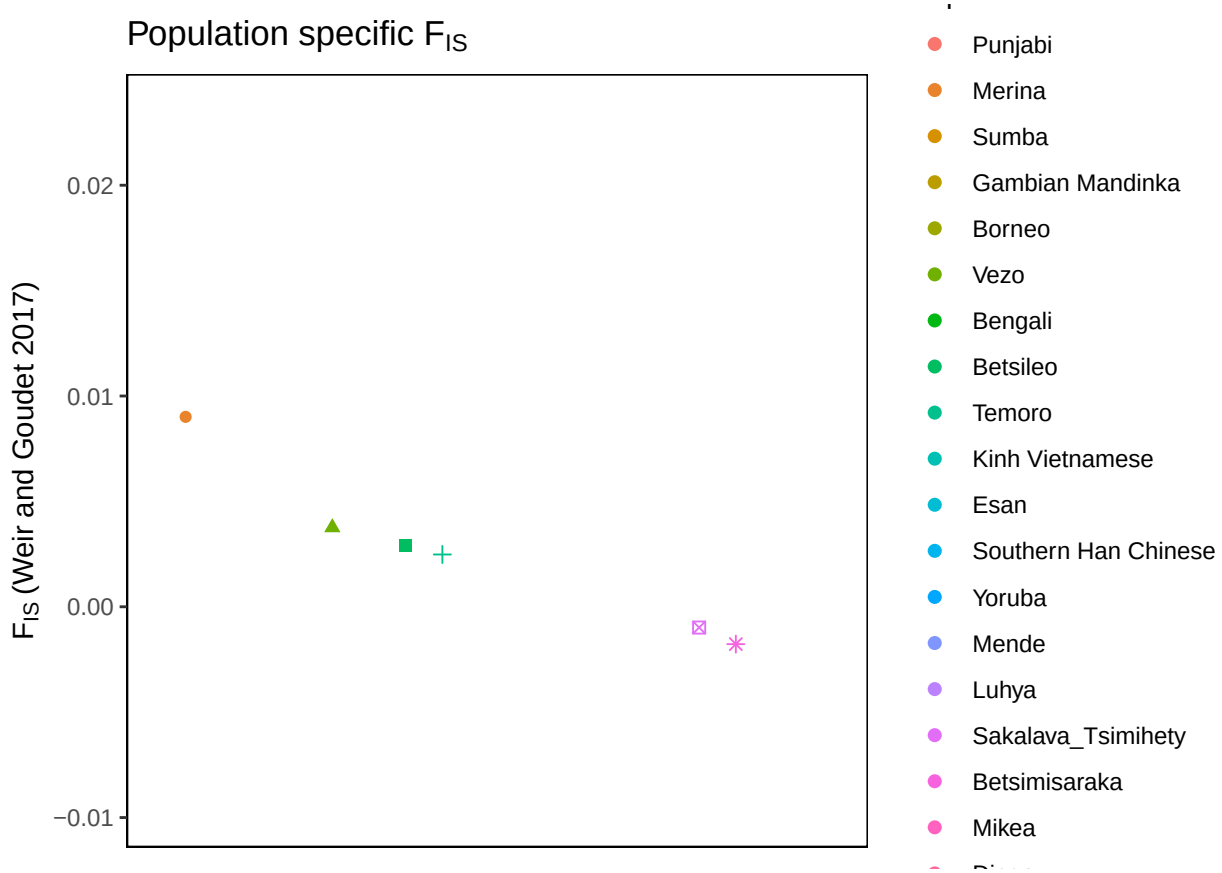

fis_all$pop_soc <- paste0(fis_all$population, "_", fis_all$social_group)
levels <- fis_all$pop_soc
levels <- levels[which(!levels %in% c("Merina_Social Group A: Merina", "Merina_Social Group B: Merina"))]
# remove everything after _ in levels
levels <- sub("_.*", "", levels)
levels <- sub("Sakalava", "Sakalava_Tsimihety", levels)
levels <- factor(levels)

fis_all$population <- factor(fis_all$population,
  levels = levels
)

ggplot(fis_all, aes(
  x = population, y = fis, fill = population,
  colour = population, shape = social_group
)) +
  geom_point(size = 1.8) +
  #scale_shape_manual(values = c(rep(21, 8), 17, 17, rep(21, 11))) +
  labs(
    title = expression("Population specific F"[IS]), x = NULL,
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)")
  ) +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank()) +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = "none"
  )+
  theme(
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```

```
## Warning: The shape palette can deal with a maximum of 6 discrete values because more
## than 6 becomes difficult to discriminate
## i you have requested 22 values. Consider specifying shapes manually if you need
##   that many have them.
```

```
## Warning: Removed 16 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggsave(paste0(res_dir, "pop_fis_global_comparison_social_groups.jpg"), width = 7, height = 7, dpi = 300)
```

```
## Warning: The shape palette can deal with a maximum of 6 discrete values because more
## than 6 becomes difficult to discriminate
## i you have requested 22 values. Consider specifying shapes manually if you need
##   that many have them.
## Removed 16 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Here, we can see that Merina Social group B has a higher F_{IS} value than Social group A, grouping closely with Mikea, Diego, and Betsimisaraka, while Social group A drops drastically below these groups. This suggests that within social group B, homozygosity is not high compared to neutral expectations.

Inspect sample sizes

```
table(malagasy$population)
```

```
##
##           Betsileo      Betsimisaraka              Diego             Merina
##                 36                 34                 31                 34
##              Mikea Sakalava_Tsimihety            Temoro              Vezo
##                 18                 32                 24                 24
```

```
table(merina$group)
```

```
##
## Social Group A: Merina Social Group B: Merina
```

```
##                 23                 11
```

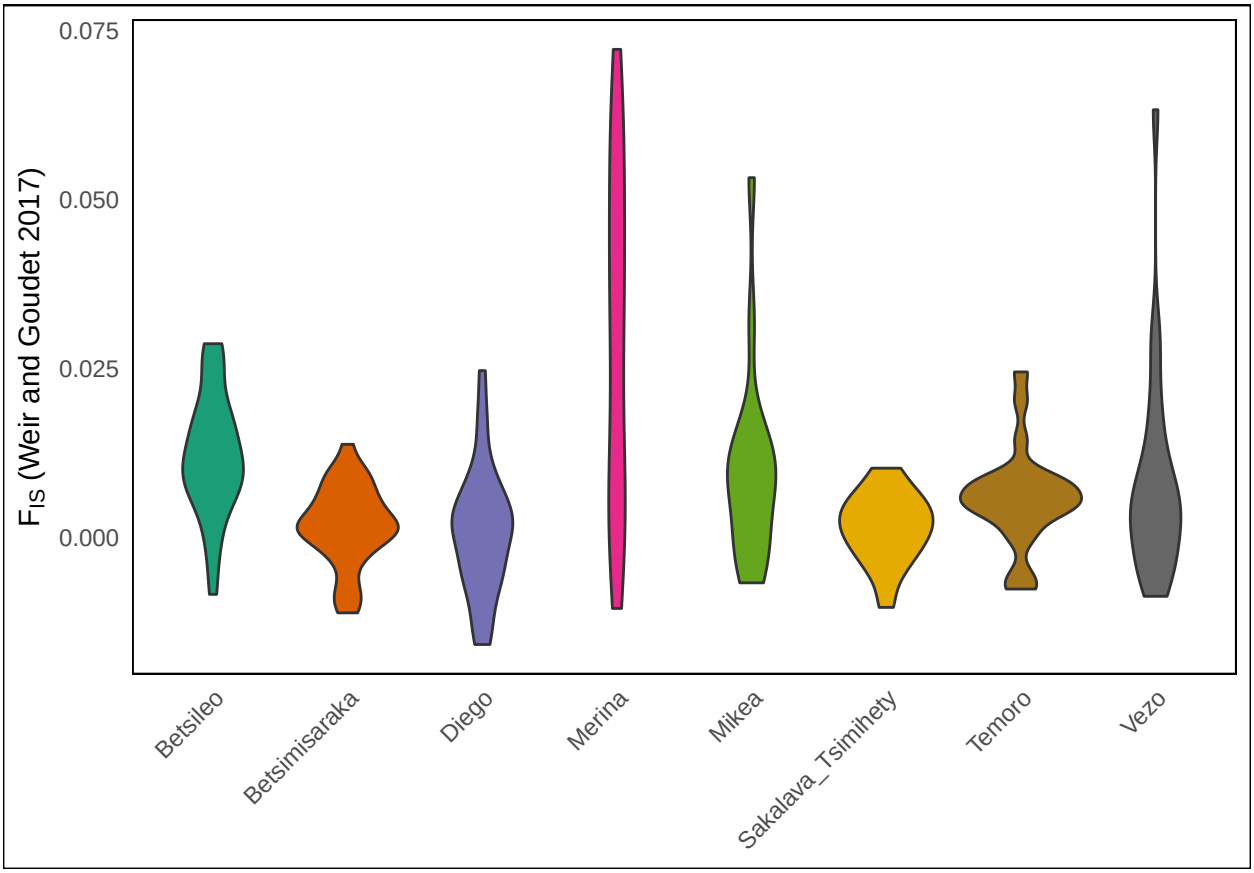
Sample size may be having an impact on F_{IS} estimates, as the sample size of Social group A is much lower than that of Social group B.

Individual F_{IS}

Subsequently, F_{IS} was calculated for each individual, also using the method of (Weir and Goudet 2017).

```
malagasy <- malagasy %>% ungroup()
malagasy <- malagasy %>% mutate(fis_indiv = indiv_inbreeding(malagasy,
  method = "WG17"
))
```

```
# plot $F for each individual IID coloured by FID
ggplot(malagasy, aes(x = population, y = fis_indiv)) +
  geom_violin(aes(fill = population)) +
  labs(
    x = NULL, y = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
    fill = "Population"
  ) +
  scale_fill_brewer(palette = "Dark2") +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank()
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none",
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```



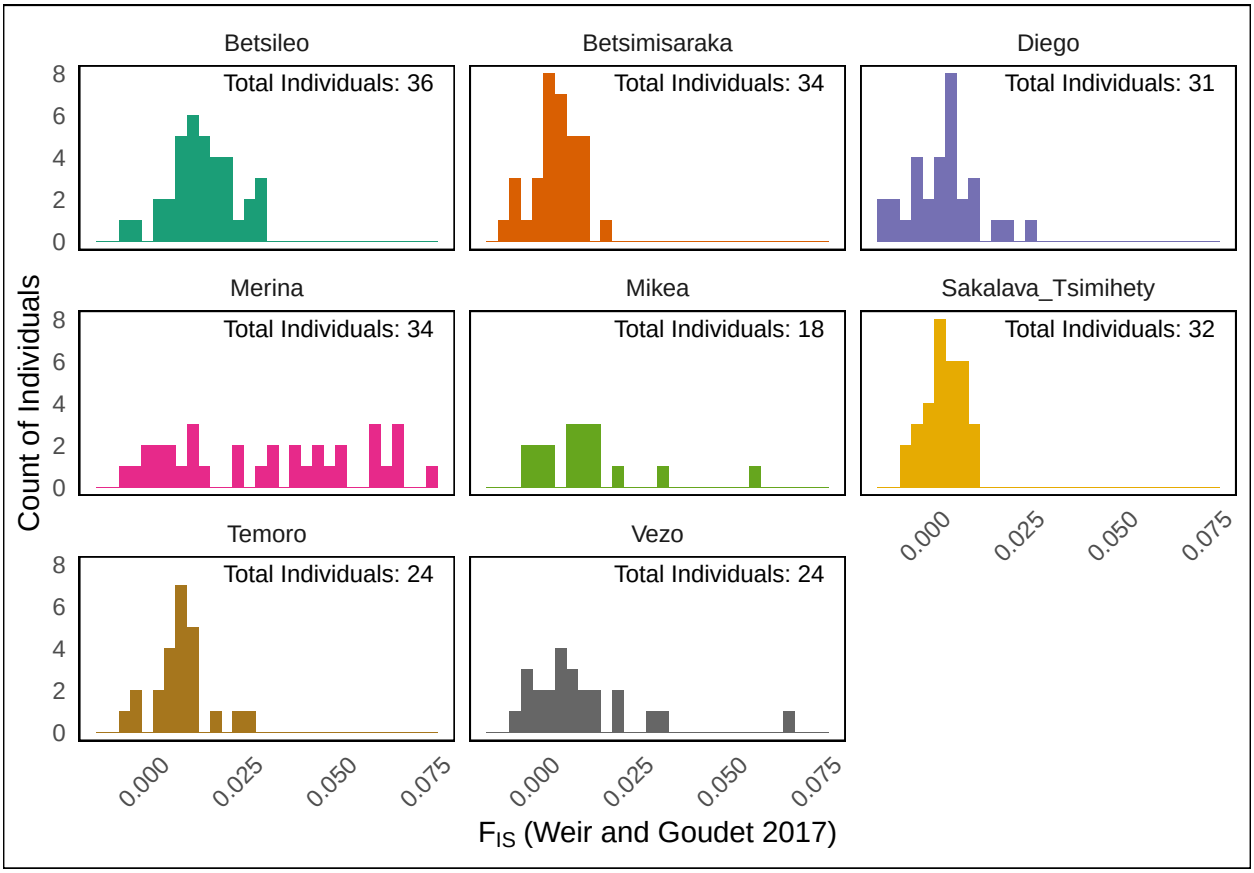

```
ggsave(paste0(res_dir, "indiv_fis_grouped.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
# create a column $total_count to count the number of individuals in each group
total_count <- table(malagasy$population)
malagasy$count <- total_count[malagasy$population]

# add total_count number to the top right corner of each facet plot
ggplot(malagasy, aes(x = fis_indiv, fill = population)) +
  geom_histogram() +
  facet_wrap(~population) +
  labs(
    y = "Count of Individuals",
    x = expression("F"[IS] ~ "(Weir and Goudet 2017)")
  ) +
  guides(fill = "none") +
  scale_fill_brewer(palette = "Dark2") +
  geom_text(
    data = malagasy %>%
      distinct(population, count), # Extract unique population and count
    aes(x = Inf, y = Inf, label = paste0("Total Individuals: ", count)),
    size = 3, hjust = 1.1, vjust = 1.5, inherit.aes = FALSE
  ) +
  theme_minimal() +
  theme(
```

```
    axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none",
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
ggsave(paste0(res_dir, "indiv_fis_hist.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

Run glm

```
glm(malagasy$fis_indiv ~ malagasy$population)
```

```
##
## Call:  glm(formula = malagasy$fis_indiv ~ malagasy$population)
##
## Coefficients:
##                             (Intercept)       malagasy$populationBetsimisaraka
```

```
##                               0.012042                               -0.010268
##              malagasy$populationDiego              malagasy$populationMerina
##                              -0.011197                                0.017319
##              malagasy$populationMikea  malagasy$populationSakalava_Tsimihety
##                              -0.002321                               -0.010548
##             malagasy$populationTemoro                malagasy$populationVezo
##                              -0.006286                               -0.003941
##
## Degrees of Freedom: 232 Total (i.e. Null);  225 Residual
## Null Deviance:       0.05819
## Residual Deviance: 0.03784   AIC: -1354
```

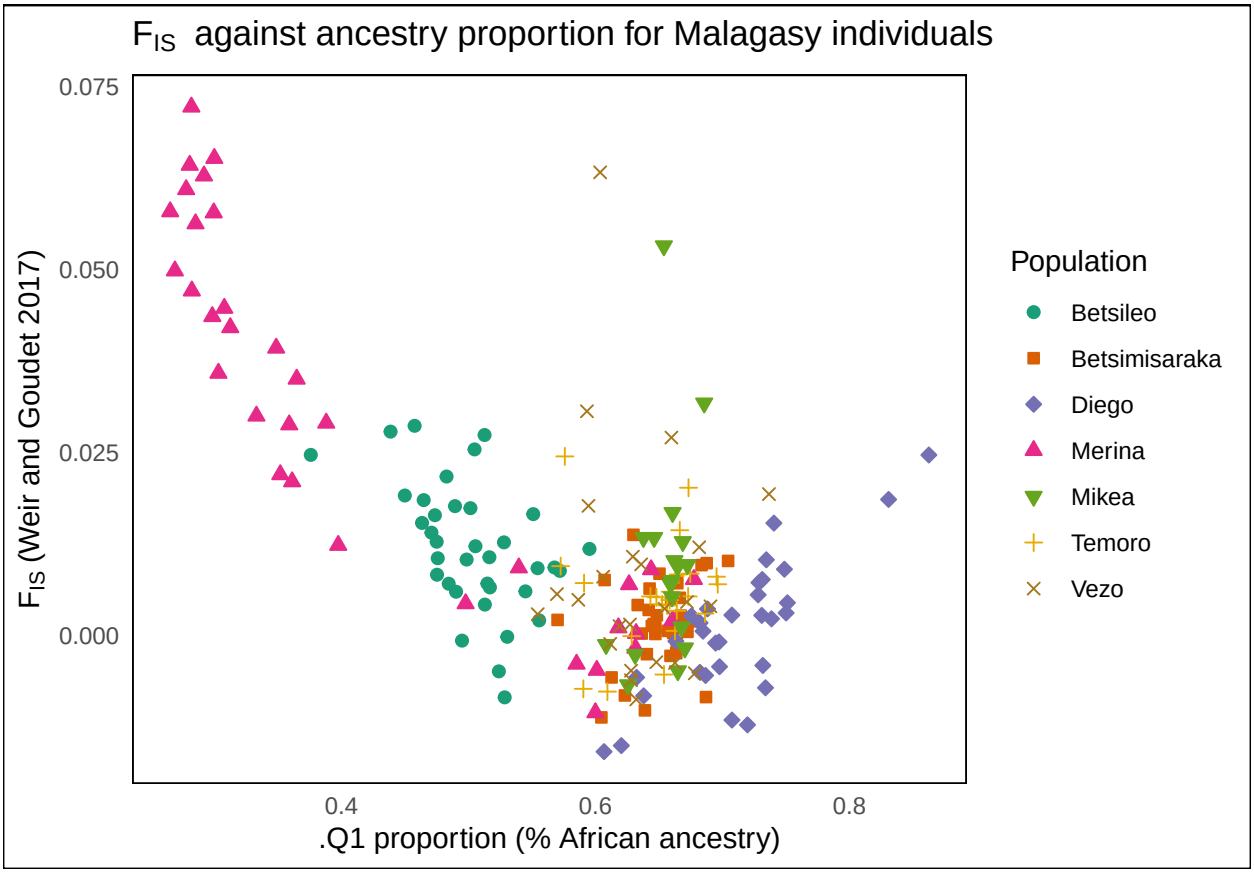
```r
# subset malagasy to individuals in malagasyq
malagasy <- malagasy %>% filter(id %in% malagasyq$id)


expect_true(identical(malagasy$id, malagasyq$id))
expect_true(identical(malagasy$population, malagasyq$population))

malagasy$.Q1 <- malagasyq$.Q1
malagasy$.Q2 <- malagasyq$.Q2
# malagasy$.Q3 <- malagasyq$.Q3

# now plot proportions vs inbreed_df$F
ggplot(malagasy, aes(
  x = .Q1, y = fis_indiv, fill = population,
  shape = population, color = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "F"[IS] ~ " against ancestry proportion for Malagasy individuals",
    x = ".Q1 proportion (% African ancestry)",
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
  ) +
  scale_fill_brewer(palette = "Dark2") +
  scale_color_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
    shape = guide_legend(title = "Population")
  ) +
  theme_minimal() +
  theme(
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```




```
ggsave(paste0(res_dir, "indiv_fis_vs_q_proportion.jpg"),
  width = 7, height = 7, dpi = 300
)
```

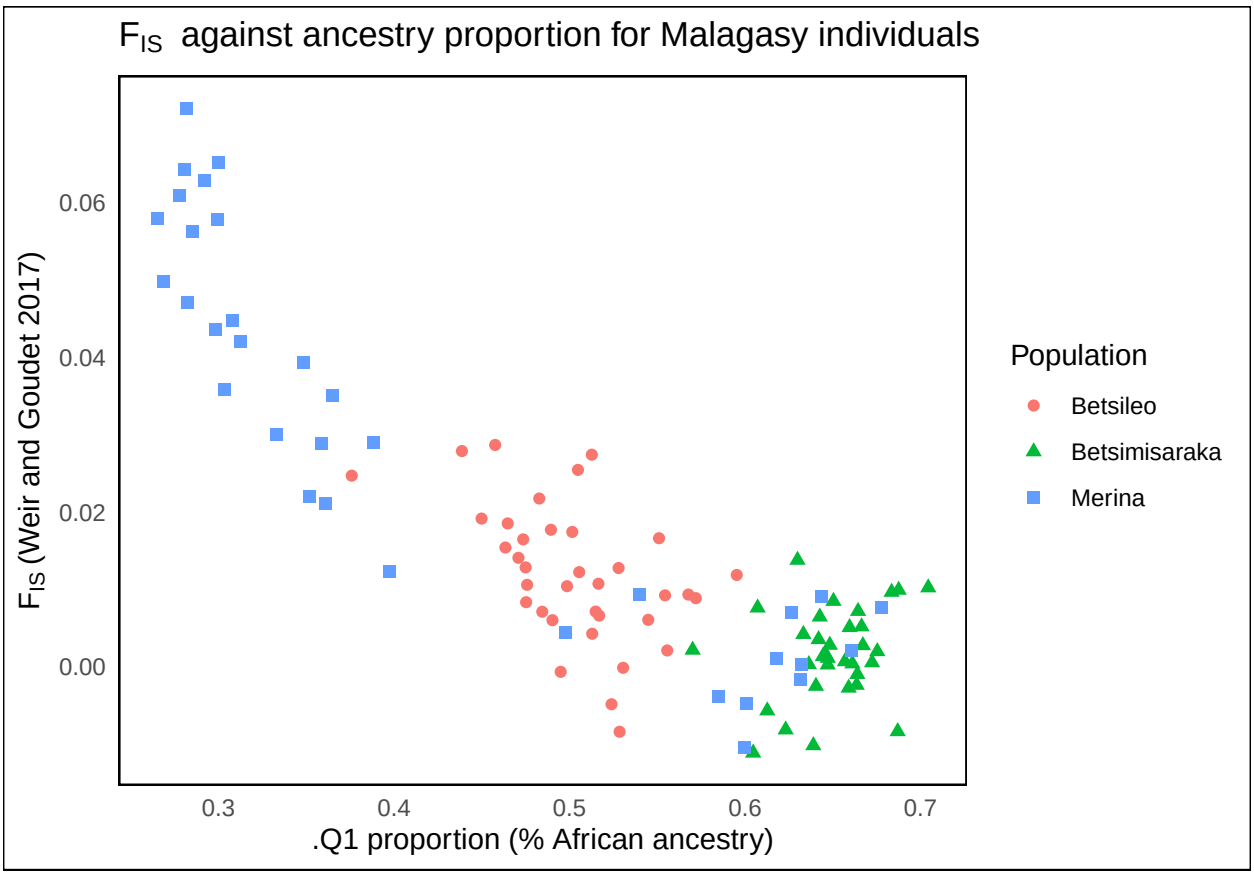
```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

```
malagasy_subset <- malagasy[malagasy$population %in% c("Betsileo", "Merina", "Betsimisaraka", "Sakalava

# now plot proportions vs inbreed_df$F
ggplot(malagasy_subset, aes(
  x = .Q1, y = fis_indiv, fill = population,
  shape = population, color = population
)) +
  geom_point(size = 1.8) +
  labs(
    title = "F"[IS] ~ " against ancestry proportion for Malagasy individuals",
    x = ".Q1 proportion (% African ancestry)",
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
  ) +
  #scale_fill_brewer(palette = "Dark2") +
  #scale_color_brewer(palette = "Dark2") +
  #scale_shape_manual(values = c(21, 22, 23, 24, 25, 3, 4, 8)) +
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank()) +
  guides(
    fill = guide_legend(title = "Population"),
    colour = guide_legend(title = "Population"),
```

```
    shape = guide_legend(title = "Population")
  ) +
  theme_minimal() +
  theme(
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```




```
ggsave(paste0(res_dir, "indiv_fis_vs_q_proportion_subset.jpg"),
  width = 7, height = 7, dpi = 300
)
```

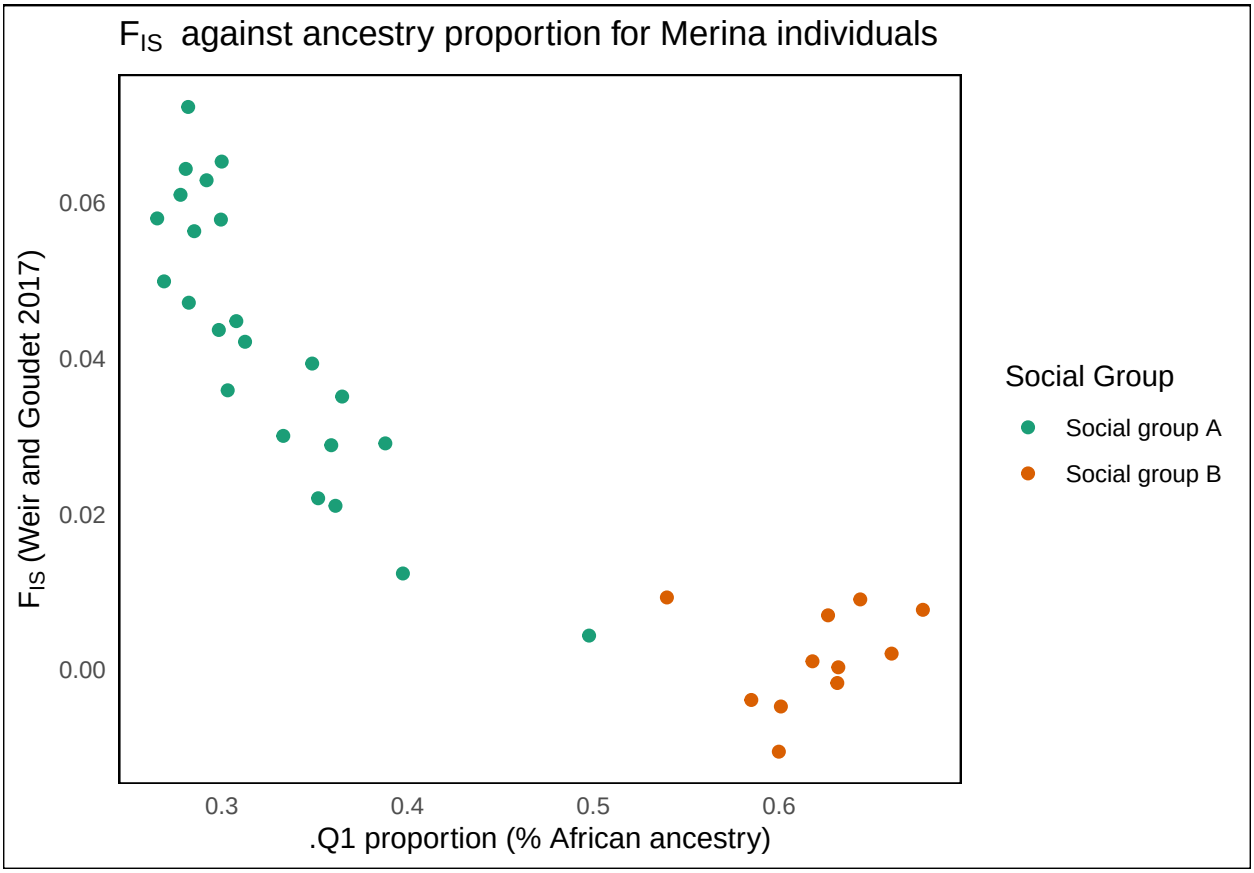
```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

```
# subset the malagasyq dataframe to only the "Merina"
merina <- malagasy %>% filter(population == "Merina")

# create a factor
merina$group <- ifelse(merina$.Q2 > 0.5, "Social group A", "Social group B")
```

```r
# plot merina$.Q3 vs merina$F
ggplot(merina, aes(x = .Q1, y = fis_indiv, fill = group, color = group)) +
  geom_point(size = 1.8, shape = 21) +
  labs(
    title = "F"[IS] ~ " against ancestry proportion for Merina individuals",
    x = ".Q1 proportion (% African ancestry)",
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
    fill = "Social Group", color = "Social Group"
  ) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
  scale_shape_manual(values = c(21, 22)) +
  theme(
    axis.ticks.x = element_blank(), axis.title.x = element_text(size = 16),
    axis.title.y = element_text(size = 16), axis.ticks.y = element_blank(),
    plot.title = element_text(size = 16),
    legend.title = element_text(size = 16),
    legend.text = element_text(size = 13)
  ) +
  theme_minimal() +
  theme(
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    plot.background = element_rect(fill = "white")
  )
```

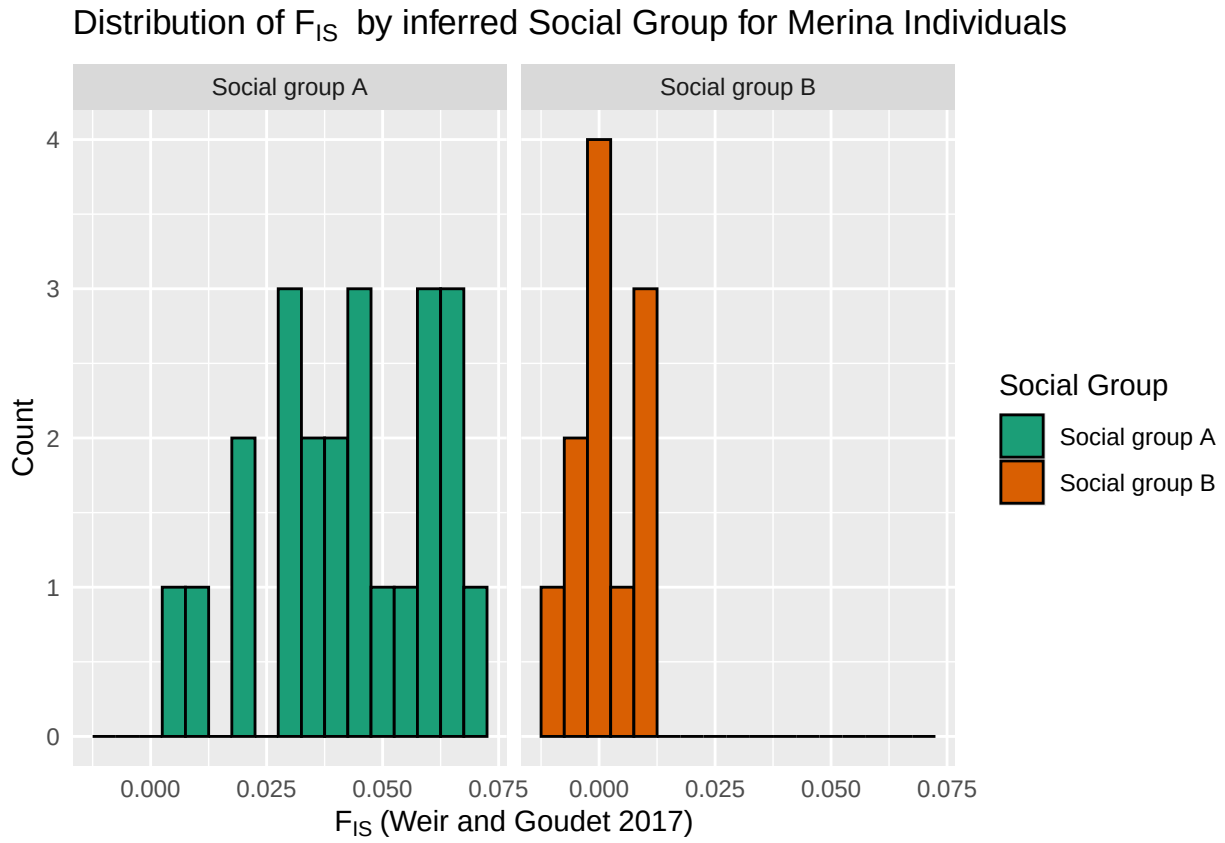
```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```




```
ggsave(paste0(res_dir, "indiv_fis_vs_q3_merina.jpg"),
  width = 7, height = 7, dpi = 300
)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

```
ggplot(merina, aes(x = fis_indiv, fill = group)) +
  geom_histogram(binwidth = 0.005, color = "black") +
  labs(
    title = "Distribution of F"[IS] ~ " by inferred Social Group for Merina Individuals",
    x = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
    y = "Count",
    fill = "Social Group"
  ) +
  scale_fill_brewer(palette = "Dark2") +
  facet_wrap(~group) + # Creates separate histograms for each 'group'
  theme(axis.ticks.x = element_blank(), axis.ticks.y = element_blank())
```




```
ggsave(paste0(res_dir, "indiv_fis_hist_merina_social_group.jpg"),
  width = 7, height = 7, dpi = 300
)
```

Run glm

```
merina$population <- factor(merina$group)

glm(merina$fis_indiv ~ merina$group)
```

```
##
## Call:  glm(formula = merina$fis_indiv ~ merina$group)
##
## Coefficients:
##                 (Intercept)  merina$groupSocial group B
##                     0.04272                    -0.04128
##
## Degrees of Freedom: 33 Total (i.e. Null);  32 Residual
## Null Deviance:       0.02033
## Residual Deviance: 0.007645  AIC: -183.1
```

Recalculate individual F_{IS} in tidypopgen, this time separating the inferred social groups

```
# subset malagasy to those with population "Merina"
merina <- malagasy %>% filter(population == "Merina")

# create a factor
merina$group <- ifelse(merina$.Q2 > 0.5, "Social group A", "Social group B")

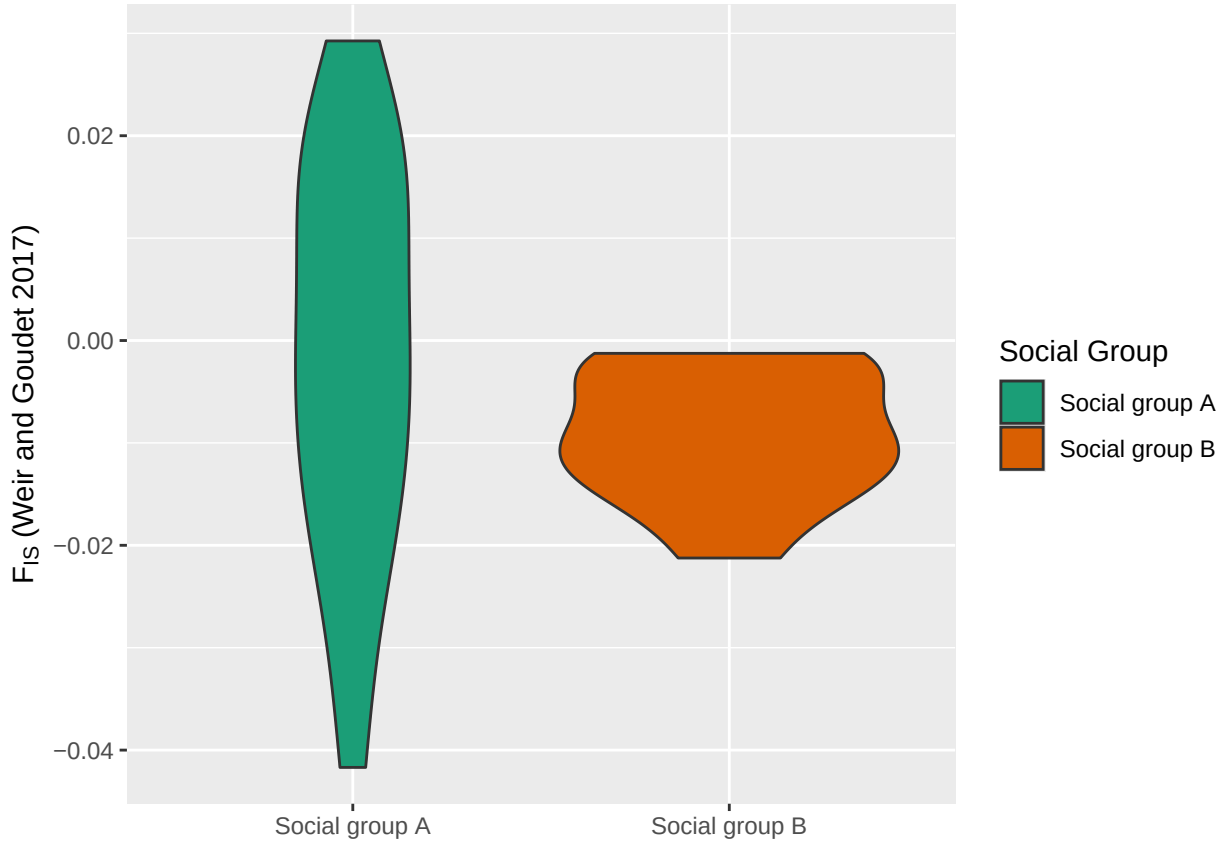
merina <- merina %>% group_by(group)

merina_fis_indiv <- merina %>% indiv_inbreeding(method = "WG17")


merina_fis_indiv <- unlist(merina_fis_indiv)
merina_fis_indiv <- as.data.frame(merina_fis_indiv)
merina_fis_indiv$id <- rownames(merina_fis_indiv)

merina <- left_join(merina, merina_fis_indiv, by = "id")
```

```
ggplot(merina, aes(x = group, y = merina_fis_indiv)) +
  geom_violin(aes(fill = group)) +
  labs(
    y = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
    x = NULL, fill = "Social Group"
  ) +
  scale_fill_brewer(palette = "Dark2")
```




```
ggsave(paste0(res_dir, "indiv_fis_merina_grouped.jpg"),
  width = 7, height = 7, dpi = 300
)
```

Individual F_{IS} for Punjabi population in 1000 Genomes

```
punjabi <- mad_bor_g1000_sum %>% filter(population == "Punjabi")
punjabi <- punjabi %>% mutate(fis_indiv = indiv_inbreeding(punjabi,
  method = "WG17"
))

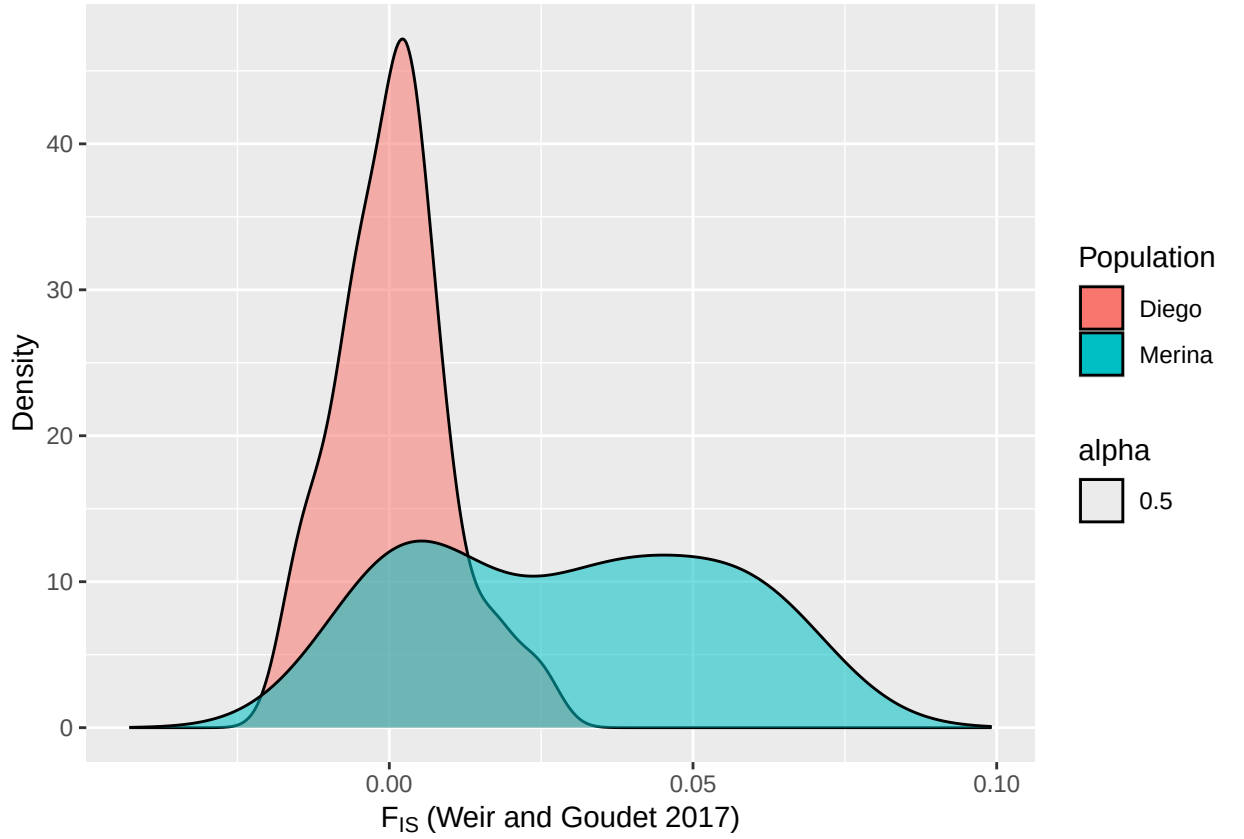
malagasy <- malagasy[, c(1:4)]

fis_combined <- rbind(malagasy, punjabi, quiet = TRUE)
fis_combined <-
  fis_combined %>% filter(population %in% c("PJL", "Merina", "Diego"))

d <- density(fis_combined$fis_indiv)

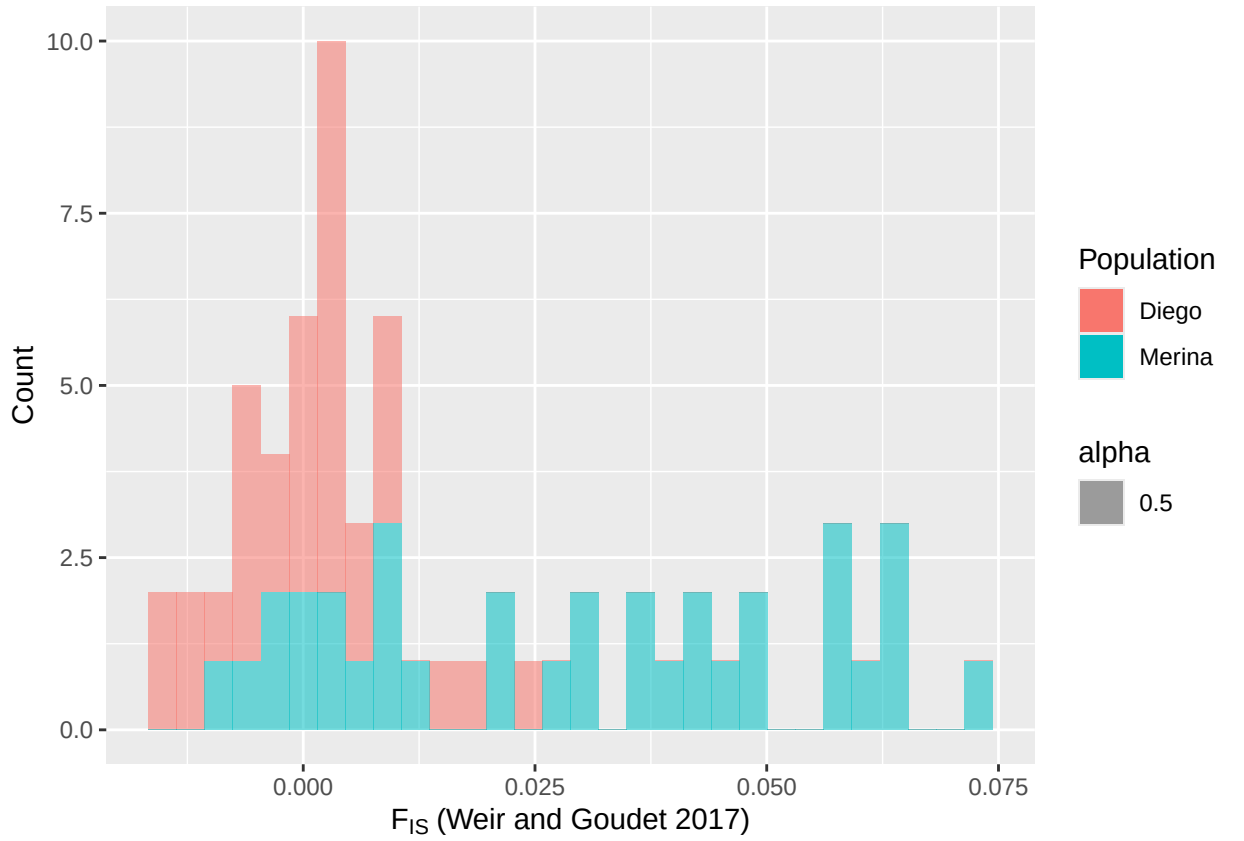
ggplot(fis_combined, aes(x = fis_indiv, fill = population, alpha = 0.5)) +
  geom_density() +
  xlim(range(d$x)) +
  labs(
    x = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
    y = "Density", fill = "Population"
```

```
  )
```




```
ggplot(fis_combined, aes(x = fis_indiv, fill = population, alpha = 0.5)) +
  geom_histogram() +
  labs(
    x = expression("F"[IS] ~ "(Weir and Goudet 2017)"),
    y = "Count", fill = "Population"
  )
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Fis for Merina compared to Punjabi, with Diego as a reference non-endogamous population. As expected, Pubjabi population has a few individuals with particularly high Fis.

Weir, Bruce S, and Jérôme Goudet. 2017. "A Unified Characterization of Population Structure and Relatedness." *Genetics* 206 (4): 2085–2103. https://doi.org/10.1534/genetics.116.198424.